Data Cleaning and Handling Inconsistencies

The dataset contained some inconsistencies and missing values, so I performed a few cleaning steps before starting the analysis.

First, the `Date` column had mixed formats such as DD-MM-YYYY and YYYY/MM/DD. I converted all values into a consistent datetime format so that time-based analysis could be done correctly.

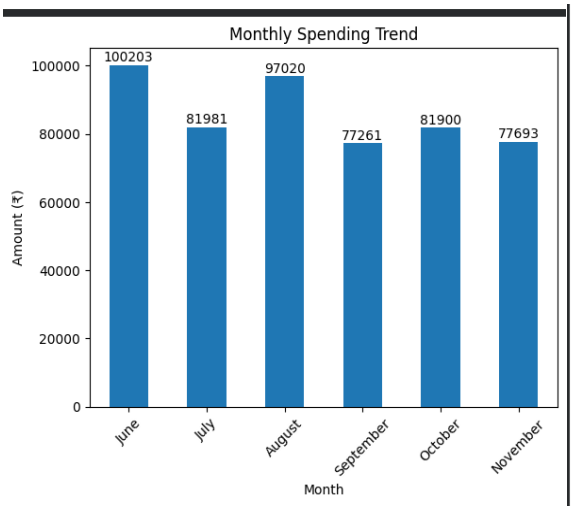
Next, the `Category` column had some missing values. Instead of removing those rows, I replaced the missing values with “Unknown” so that no data was lost.

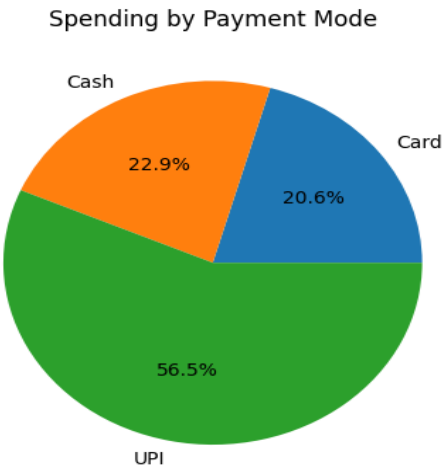
I also noticed that some category names were inconsistent, like “food” and “Food”. I standardized them by converting all category names to a uniform format.

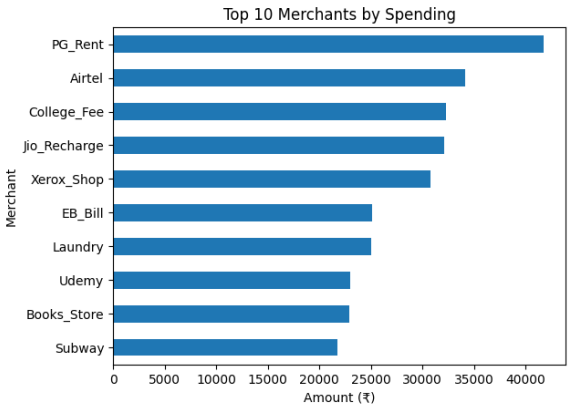
Additionally, I checked for duplicate records and ensured the dataset was clean.

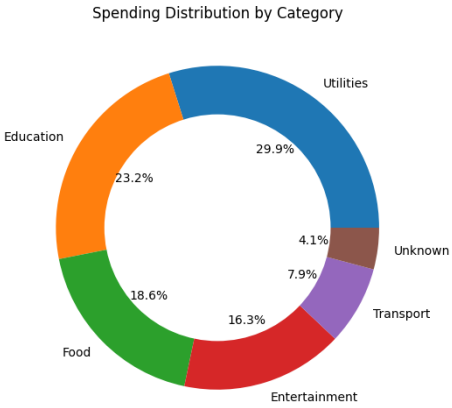
Finally, I created new columns such as `Day` and `Month` from the Date column to help analyze spending patterns across different time periods

Visualizations and Key Insights









- The **monthly spending chart** shows that expenses are highest in June and August, while other months remain fairly stable, indicating some one-time high expenses followed by consistent spending.
- The **payment method chart** shows that UPI accounts for more than half of the transactions, meaning the student mainly uses digital payments.
- The **top merchants chart** highlights that most money is spent on rent, bills, and essential services, showing that a large portion of expenses is fixed.
- The **category distribution chart** shows that Utilities and Education take up the biggest share, while Food and Entertainment also contribute significantly.

Summary

The analysis shows that most of the student's spending goes towards essential categories like Utilities and Education. Spending is higher in certain months due to one-time expenses, but remains stable overall. Weekend and lifestyle expenses like Food and Entertainment also contribute significantly. Overall, while basic expenses are fixed, controlling discretionary spending can help in better budget management.

Budget Recommendation

Based on the analysis, the student should focus on controlling spending during weekends, especially in categories like Food and Entertainment. Setting a weekly limit for these expenses and tracking high-value transactions can help in reducing unnecessary spending and improving savings.